

Optimal irrationality measures for logarithms of rational numbers and a compatible-embedding extension

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Abstract

We prove that the irrationality exponent of $\log r$ is 2 for every positive rational number $r \neq 1$, in particular for $r = 3$. The proof extends the separated-weight interpolation and determinant method of OpenAI's proof that the irrationality exponent of π is two. The geometric extension permits distinct nonzero multiplicative centers, and the arithmetic extension accounts for denominators of rational and algebraic bases. More generally, let F be a number field of degree d , let $\gamma, \xi \in F$ with $\gamma \neq 0$, and let $x \in \mathbb{R} \setminus \{0\}$. If $\exp(\sigma(\gamma)x) = \sigma(\xi)$ for $s \geq 1$ embeddings $\sigma : F \hookrightarrow \mathbb{C}$, then x is irrational and $2 \leq \mu(x) \leq 2d/s$. The same algebraic determinant is estimated at every embedding; the field norm supplies the factor d/s . The root-of-unity case is included. Consequences include a bound $2 \deg_{\mathbb{Q}}(\alpha)$ for $\mu(\log \alpha)$ when $\alpha > 0$ is algebraic and not 1, and exponent two for rational-power logarithms, nonzero rational arctangents, and certain quadratic-normalized logarithms. The endpoint statements and their full dependency closure have been checked in Lean, including replay in an initially empty kernel environment. The proof does not compute numerical denominator thresholds.

Keywords: irrationality exponent; logarithms; weighted interpolation; algebraic number fields; formal verification.

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1 Introduction

For an irrational real number x , write

$$\mu(x) = \sup \left\{ \nu > 0 : 0 < \left| x - \frac{p}{q} \right| < q^{-\nu} \text{ for infinitely many reduced } p/q, q \geq 2 \right\},$$

allowing the value $+\infty$. Dirichlet's theorem gives $\mu(x) \geq 2$. An upper bound concerns every sufficiently large denominator, not merely a chosen sequence of approximants. All logarithms of positive real numbers in this paper are natural.

Theorem 1.1 (Rational logarithms). *Let $r \in \mathbb{Q}$, $r > 0$, $r \neq 1$. For every $\varepsilon > 0$ there is an integer $Q = Q(r, \varepsilon) \geq 2$ such that*

$$\left| \log r - \frac{p}{q} \right| \geq q^{-(2+\varepsilon)} \quad (p \in \mathbb{Z}, q \in \mathbb{N}, q \geq Q). \quad (1.1)$$

No coprimality condition is imposed. Consequently $\log r$ is irrational and

$$\mu(\log r) = 2.$$

In particular, $\mu(\log 3) = 2$.

The broader statement records exactly how the number field enters. Embeddings, including two complex-conjugate embeddings, are counted separately.

Theorem 1.2 (Compatible embeddings). *Let F be a number field of degree $d = [F : \mathbb{Q}]$, let $\gamma, \xi \in F$, $\gamma \neq 0$, and let $x \in \mathbb{R} \setminus \{0\}$. Suppose that a nonempty set S of embeddings $F \hookrightarrow \mathbb{C}$ satisfies*

$$\exp(\sigma(\gamma)x) = \sigma(\xi) \quad (\sigma \in S). \quad (1.2)$$

Put $s = |S|$ and $\kappa = d/s$. For every $\varepsilon > 0$ there is an integer $Q \geq 2$ such that

$$\left| x - \frac{p}{q} \right| \geq q^{-(2\kappa+\varepsilon)} \quad (p \in \mathbb{Z}, q \in \mathbb{N}, q \geq Q). \quad (1.3)$$

In particular x is irrational and

$$2 \leq \mu(x) \leq \frac{2d}{s}.$$

If all embeddings are compatible, then $\mu(x) = 2$.

Compatibility already implies $\xi \neq 0$. There is no hypothesis that ξ is not a root of unity, that x is already irrational, or that an interpolation matrix has full rank. Those facts or alternatives needed in the proof are established below.

Corollary 1.3 (Positive algebraic bases). *If $\alpha > 0$ is algebraic over $\mathbb{Q}$, $\alpha \neq 1$, and $d = \deg_{\mathbb{Q}}(\alpha)$, then $\log \alpha$ is irrational and*

$$2 \leq \mu(\log \alpha) \leq 2d.$$

For every $\varepsilon > 0$, the associated lower bound with exponent $2d + \varepsilon$ holds for all sufficiently large denominators.

Proof. Take $F = \mathbb{Q}(\alpha) \subset \mathbb{R}$, $\gamma = 1$, $\xi = \alpha$, $x = \log \alpha$, and let S contain its given real embedding into $\mathbb{C}$. Theorem 1.1 is the case $F = \mathbb{Q}$, $s = d = 1$. $\square$

1.1 Prior work and the contribution here

Quantitative approximation to logarithms has a substantial history. Zudilin’s survey [12] discusses integral and hypergeometric methods for $\log 2$, $\log 3$, and π . For $\log 3$, Wu and Wang [11, Corollary 1] obtained $\mu(\log 3) < 5.1163051$, subsequently improved by Bondareva, Luchin, and Salikhov [1, Corollary 1] to $\mu(\log 3) \leq 5.116201$. These are finite upper bounds, not the exact value two asserted in Theorem 1.1. Marcovecchio and Viola [6] study approximation to logarithms by elements of number fields, as well as nonquadraticity measures, using double integrals and the saddle-point method. Their number-field approximation problem, normalized by Weil height, must be distinguished from the rational-approximation exponent used here. Marcovecchio’s multiple-Legendre construction [5] also treats approximation by algebraic numbers of bounded degree. Bouchelaghem, He, Li, and Wu [2] study linear-independence measures involving two rational logarithms, another quantitative problem distinct from the individual exponents considered here. We do not claim that finite irrationality bounds for logarithms were previously unknown, nor that our degree bound is best in every special case.

The rational-logarithm theorem was obtained independently and contemporaneously with Liu, Jiang, and Zhang [4]. We became aware of their preprint after completing our work. The present paper also establishes the compatible-embedding theorem and its additional applications. In particular, Theorem 1.2 controls a single algebraic determinant at all embeddings and obtains the factor d/s by taking its field norm; the results include the degree bound of Corollary 1.3, rational arctangents, and quadratic-normalized logarithms.

The immediate input is OpenAI’s preprint [8] and its pinned formal development. Its proof of $\mu(\pi) = 2$ combines separated approximation denominators, weighted interpolation, arithmetic clearing of a nonzero determinant, and an analytic saving caused by repeated Taylor coefficient rows. This paper reuses that architecture and its proved general ingredients. It is not a deduction from the numerical statement $\mu(\pi) = 2$.

The moving-center geometric input allows the multiplicative coordinates of the interpolation points to be distinct nonzero numbers. For $\log 3$ they are 3^j ; for $\log r$, they are r^j . The local coordinate $t = Y/y_j - 1$ preserves the logarithmic frame. Once the rigidity argument forces Y to be constant, such a curve can meet at most one center. The formal development also supplies the common-fiber alternative needed when a root of unity is reduced to 1. Both alternatives lead to the actual, explicitly defined interpolation matrix, not to a matrix of assumed rank.

The arithmetic extension clears denominators of both the algebraic base ξ and the slope γ . A single selected determinant over F is then mapped through every embedding. Compatible embeddings give analytic decay; the others incur a controlled growth cost. Taking its norm and dividing by the number of compatible embeddings produces $\kappa = d/s$. Omitting the other embeddings would erase this essential factor and would not prove the stated theorem.

1.2 Organization and scope

Section 2 fixes the approximation conventions. Section 3 proves the required interpolation statement, making the inherited multiplicity input explicit. Sections 4 and 5 establish the arithmetic and analytic inequalities for the same determinant. Section 6 chooses all parameters in their required order and closes the contradiction. Section 7 gives the specializations, a comparison with the π proof, and counterexamples to unrestricted real-input claims. Section 8 records formal statements, source provenance, and reproducible checks.

The results do not assert exponent two for every positive algebraic base or a uniform finite bound for arbitrary positive real bases. The eventual thresholds are existential; this paper does not

compute their numerical values.

2 Approximation conventions and preliminary reductions

Definition 2.1. For $x, b \in \mathbb{R}$, write $\text{ELB}(x, b)$ if, for every real $\nu > b$, there is $Q \in \mathbb{N}$, $Q \geq 2$, such that

$$\forall p \in \mathbb{Z} \forall q \in \mathbb{N}, \quad q \geq Q \implies q^{-\nu} \leq |x - p/q|.$$

Lemma 2.2. If $\text{ELB}(x, b)$, then x is irrational. Moreover, for $b \geq 2$,

$$2 \leq \mu(x) \leq b.$$

Proof. If $x = a/c$, with $a \in \mathbb{Z}$, $c \in \mathbb{N}_{>0}$, use $p = ka$, $q = kc$ with k arbitrarily large. This contradicts the positive lower bound in the definition. For any $\nu > b$, that definition excludes good reduced approximants with $q \geq Q$. There are only finitely many smaller denominators and finitely many relevant numerators for each of them. Thus no approximation exponent greater than b belongs to the defining set of $\mu(x)$. Dirichlet supplies the opposite inequality. $\square$

It is enough to contradict the following assertion for each fixed $\nu > 2\kappa$:

$$\text{for every } Q, \quad \exists p \in \mathbb{Z}, q \in \mathbb{N}, \quad q \geq Q, \quad |x - p/q| \leq q^{-\nu}. \quad (2.1)$$

Failure of $\text{ELB}(x, 2\kappa)$ implies this assertion for some such ν . Since $x \neq 0$, sufficiently large approximants in (2.1) have $p \neq 0$. They also satisfy $|p/q| \leq |x| + 1$. The integers

$$w_i = \lceil \log q_i \rceil$$

can be made successively as large as desired. In particular, one may require each w_i to exceed a threshold depending on the previously chosen weights. This does not require a prescribed growth rate of the whole sequence of good approximants.

Lemma 2.3 (Root-of-unity reduction). *In proving Theorem 1.2, one may assume either that the powers ξ^j , $j \geq 0$, are distinct or that $\xi = 1$. The reduction preserves x, F, S, d, s .*

Proof. If ξ is not a root of unity, equality of two distinct powers would give a positive power equal to 1, since $\xi \neq 0$. If $\xi^n = 1$ for some $n \geq 1$, replace (γ, ξ) by $(n\gamma, 1)$. Equation (1.2) remains true after raising both sides to the n -th power. The new slope is nonzero. $\square$

From now through Section 6, this normalization has been performed. Its effect is only on fixed constants. All choices other than the final degree parameter H are fixed before any limit $H \rightarrow \infty$ is taken. For multi-indices $\alpha \in \mathbb{N}^m$, put

$$|\alpha| = \sum_{i=1}^m \alpha_i, \quad w\alpha = \sum_{i=1}^m w_i \alpha_i, \quad \binom{\alpha}{\beta} = \prod_{i=1}^m \binom{\alpha_i}{\beta_i}.$$

Coordinatewise inequalities between multi-indices have their usual meaning. Empty products and empty least common multiples equal 1.

3 Interpolation at logarithmic centers

Let $m, K \geq 1$, and fix positive rational w_0, v_0, θ with $0 < \theta < 1$. For further positive rational weights w_i , set

$$W = (w_0, w_1, \dots, w_m), \quad V = (v_0, v_1, \dots, v_m), \quad v_i = w_i/\theta \quad (i > 0).$$

Let $\mathcal{P}_W(H)$ be the space spanned by $Y^h X^\alpha$ with $w_0 h + w\alpha \leq H$. Let $\mathcal{J}_V(H)$ be the quotient of $\mathbb{C}[[t, u_1, \dots, u_m]]$ retaining precisely the coefficients

$$v_0 \ell + \sum_i v_i \beta_i < H.$$

The inequality for rows is strict, whereas the inequality for columns is weak.

Theorem 3.1 (Separated logarithmic interpolation). *Assume*

$$K(w_0/v_0)\theta^m < 1. \tag{3.1}$$

There are successive lower thresholds for $w_1, \dots, w_m$, independent of the center coordinates, with this property. For nonzero $y_j \in \mathbb{C}$, $0 \leq j < K$, and $c_{ji} \in \mathbb{C}$, the map

$$\begin{aligned} \mathcal{P}_W(H) &\longrightarrow \bigoplus_{j < K} \mathcal{J}_V(H), \\ P &\longmapsto (P(y_j(1+t), (c_{ji} + u_i + \log(1+t))_{i=1}^m))_{j < K} \end{aligned} \tag{3.2}$$

is surjective for all sufficiently large H divisible by a fixed positive integer, in either of the following cases:

- (i) *the y_j are pairwise distinct;*
- (ii) *all $y_j = 1$, each family $c_{0i}, \dots, c_{K-1,i}$ is pairwise distinct, and $K\theta^m < 1$.*

The divisibility requirement depends only on the weights. The lower threshold for H may depend on every fixed datum, including the centers.

Case (ii) is the common-fiber interpolation of [8, Section 2]. We give the argument with both cases present to exhibit the exact change required for case (i). In particular, a count of coefficients alone is not used to assert surjectivity.

3.1 The uniform local input and separated weights

We use the weighted transverse multiplicity comparison of [8, Lemma 2.2]. In the notation needed here it says the following.

Lemma 3.2 (Weighted multiplicity comparison). *Give D affine coordinates positive degree weights ρ_a . Let commuting analytic vector fields form a frame on an open set, with positive derivative costs κ_b . Suppose polynomials f_ℓ have weighted degree at most N , and Z is a codimension- k component of their common zero set. If every further frame derivative of cost less than ϵN of every f_ℓ vanishes on Z , then, for any coordinate normal basis A and frame normal basis B ,*

$$\prod_{a \in A} \rho_a \leq 2k! \epsilon^{-k} \prod_{b \in B} \kappa_b. \tag{3.3}$$

The constant is independent of the equations and of the component.

Proof. Work at a general smooth point of Z where both normal-basis conditions hold and no other component passes through the point. The commuting flows in the B -directions give transverse analytic coordinates. Vanishing of the derivatives says that the equations have no transverse terms of weight below ϵN . Restrict to the coordinate slice in the A -directions. That slice is transverse, its common zero is isolated, and substitution of the remaining analytic coordinates cannot introduce a transverse term of smaller weight. Thus the local quotient length is at least

$$\#\{\alpha \in \mathbb{N}^k : \sum_{b \in B} \kappa_b \alpha_b < \epsilon N\} \geq \frac{(\epsilon N)^k}{2k! \prod_{b \in B} \kappa_b}.$$

For the last inequality, unit cubes indexed by the integer parts of points of the open weighted simplex cover that simplex; even the factor 2 is unnecessary for this count.

Generic k constant linear combinations of the restricted equations still cut out an isolated zero. Their quotient length bounds the preceding length from above, and weighted Bézout bounds it by $N^k / \prod_{a \in A} \rho_a$. For clarity, this weighted version follows by choosing an integer M_0 with $M_0 \rho_a \in \mathbb{N}_{>0}$ and substituting $x_a = \lambda_a + z_a^{M_0 \rho_a}$. Choose λ_a so that the preimages of the point are unramified. There are $\prod_{a \in A} M_0 \rho_a$ such preimages, each preserving the local length, while the total degrees are at most $M_0 N$. Ordinary Bézout for isolated zeros gives the asserted upper bound; see [7, Corollary VIII.3]. Comparison of the two lengths proves (3.3). $\square$

Choose $\sigma_0 \in \mathbb{Q}_{>0}$ small enough that

$$\begin{aligned} (1 + 3\sigma_0)^{m+1} K(w_0/v_0) \theta^m &< 1, \\ (1 + \sigma_0) \theta &< 1, \end{aligned} \tag{3.4}$$

and, in case (ii), also $(1 + 3\sigma_0)^m K \theta^m < 1$. Put $L_0 = m + 2$ and take the deliberately enlarged constant

$$C_* = 2(m+2)^{m+2} \left(1 + \frac{m+2}{\sigma_0}\right)^{m+2}.$$

It dominates $2k!(L_0/\sigma_0)^k$ for $1 \leq k \leq m$. Choose the positive weights successively so that

$$\prod_{a \in A} w_a > C_* \prod_{b \in B} v_b \tag{3.5}$$

whenever $A, B \subseteq \{0, \dots, m\}$ have equal size between 1 and m , and the largest positive index in their symmetric difference belongs to A . At that index the ratio in (3.5) contains the new weight linearly. Larger common indices contribute only $w_i/v_i = \theta$; all remaining factors were already fixed. Finitely many inequalities therefore impose a finite lower threshold. No point coordinate enters these thresholds.

The formal implementation can use a rectangular transverse-length bound, with constant $k^k (L_0/\sigma_0)^k$, at a sufficiently large auxiliary degree. The displayed C_* also dominates that constant. The auxiliary-degree threshold is independent of the varying component, so either local formulation gives the same separated-weight argument.

3.2 The curve inequality

On the smooth complete model of an irreducible curve C meeting the affine chart, define

$$\deg_W C = \sum_P \max \left\{ 0, -\frac{\text{ord}_P Y}{w_0}, -\frac{\text{ord}_P X_1}{w_1}, \dots, -\frac{\text{ord}_P X_m}{w_m} \right\}. \tag{3.6}$$

An identically zero coordinate contributes no pole. At a branch through $a_j = (y_j, c_{j1}, \dots, c_{jm})$, let

$$h_P = \min \left\{ \frac{\text{ord}_P(Y/y_j - 1)}{v_0}, \frac{\text{ord}_P(X_i - c_{ji} - \log(Y/y_j))}{v_i} : 1 \leq i \leq m \right\}. \quad (3.7)$$

The logarithm is a formal series at the branch, not a global choice of a complex logarithm. These contacts are positive and finite on a nonconstant branch.

Proposition 3.3. *For the separated weights above, in either case of Theorem 3.1,*

$$\deg_W C \geq (1 + \sigma_0) \sum_{P \mapsto a_j} h_P. \quad (3.8)$$

Proof. Suppose the inequality fails. A count of coefficients and conditions gives, for large auxiliary integers N , a nonzero polynomial F_N of W -degree at most N , with local V -order at least $(1 + 3\sigma_0)N$ at every center. The leading terms in the two counts are

$$\frac{N^{m+1}}{(m+1)! \prod_a w_a}, \quad \frac{K(1 + 3\sigma_0)^{m+1} N^{m+1}}{(m+1)! \prod_a v_a},$$

and (3.4) makes the first larger. Independence of the conditions is not needed to produce F_N .

On $Y \neq 0$, use the commuting frame

$$D_0 = Y \partial_Y + \sum_{i=1}^m \partial_{X_i}, \quad D_i = \partial_{X_i} \quad (i > 0).$$

In the coordinates of (3.2), these are $(1+t)\partial_t, \partial_{u_i}$. Each preserves the bound on polynomial degree; differentiating with cost v_a lowers local weighted order by at most v_a . Thus every derivative of F_N of total cost at most $\sigma_0 N$ has branch order at least $(1 + 2\sigma_0)N h_P$. Its total pole order on the complete curve is at most $N \deg_W C$. The assumed failure of (3.8) makes the former total strictly larger. The zero–pole equality forces all these derivatives to vanish identically on C .

Set $\delta_N = \sigma_0 N / L_0$. Consider successively the common zero sets of derivatives of cost at most $r\delta_N$, $0 \leq r \leq L_0$, on $Y \neq 0$. Choose nested irreducible components containing the affine part of C , working backwards from the last level. Their dimensions lie between 1 and m ; hence two adjacent components coincide. Call this persistent component Z , of codimension k , $1 \leq k \leq m$. Further derivatives of cost less than δ_N of its defining equations from the earlier level vanish on Z . Lemma 3.2 gives

$$\prod_{a \in A} w_a \leq C_* \prod_{b \in B} v_b$$

for every coordinate normal basis A and frame normal basis B . This argument is an instance of the persistent derivative-ideal method; see also [9, Section 5].

If some positive coordinate direction is nontangent to Z , choose its largest index i . Some coordinate normal basis contains i . Separation forces every frame normal basis to contain D_i . The span of the remaining frame vectors is $\ker(dX_i - dY/Y)$. Its failure to span the normal quotient implies

$$dX_i = dY/Y \quad \text{on } Z.$$

This forces Y to be constant. Otherwise restrict to an algebraic curve on which Y is nonconstant and pass to its complete model. At a zero or pole of Y , the residue of dY/Y is the nonzero integer $\text{ord}_P Y$, whereas the exact differential dX_i has residue zero. This is impossible. If every positive

coordinate direction is tangent, properness of Z forces its tangent space to be their span, so $dY = 0$ and the same conclusion holds. In particular Y is constant on C .

In case (i), C meets at most one prescribed center a_j . Choose X_ℓ nonconstant on C . The function $X_\ell - c_{j\ell}$ has total pole degree at most $w_\ell \deg_W C$, and has zero order at least $v_\ell h_P$ at every relevant branch. Consequently

$$\deg_W C \geq \theta^{-1} \sum_{P \mapsto a_j} h_P > (1 + \sigma_0) \sum_{P \mapsto a_j} h_P,$$

contradicting the supposition.

In case (ii), the constant is $Y = 1$. In that fiber a second coefficient count is available because $K(1 + 3\sigma_0)^m \theta^m < 1$. It gives a nonzero polynomial in $X_1, \dots, X_m$ of degree at most N and order at least $(1 + 3\sigma_0)N$ at all centers. The same zero–pole argument makes its low-cost ordinary partial derivatives vanish on C . For $m = 1$, a nonzero polynomial cannot vanish on this curve. For $m \geq 2$, repeat the persistent-component argument in the fiber. The coordinate and derivative frames now agree. Two different normal-basis index sets would violate (3.5), so there is just one. Choose an index i in it. The other coordinate directions fail to span the normal quotient, so $dX_i = 0$ on the component. That coordinate is constant in characteristic zero. Coordinatewise distinctness implies that the component, and hence C , meets at most one center. The preceding single-center zero–pole argument again contradicts the supposed excess. $\square$

3.3 From curve degrees to the actual coefficient map

We finish the proof of Theorem 3.1; this is the curve-to-jet passage of [8, Section 2.3], with normalized coordinates at arbitrary $y_j \neq 0$. Choose an integer R for which every R/w_a and R/v_a is a positive integer. Embed the affine space by all monomials of W -degree at most R , including 1, and take its projective closure X , with $L = \mathcal{O}_X(1)$. Pure coordinate powers attain the maximum pole at each point of a complete curve, so

$$\deg(L|_C) = R \deg_W C. \quad (3.9)$$

Replace each logarithm by a Taylor polynomial $G_i(t)$ whose first omitted term has t -weight greater than v_i . The coordinates

$$t = Y/y_j - 1, \quad u'_i = X_i - c_{ji} - G_i(t)$$

are algebraic regular coordinates at a_j . The replacement preserves (3.7) and induces inverse filtration-preserving triangular changes on every finite coefficient quotient. Define the finite-colength ideal

$$I_j = (t^{R/v_0}, (u'_1)^{R/v_1}, \dots, (u'_m)^{R/v_m}).$$

Patch these ideals with the unit ideal away from the centers. Its branch order is Rh_P .

Let $p : X' \rightarrow X$ be the ordinary blow-up of this ideal I , and write $I\mathcal{O}_{X'} = \mathcal{O}_{X'}(-E)$, $A = p^*L$. The curve inequality and (3.9) show that $A - (1 + \sigma_0)E$ is nef. For a curve contracted by p , this follows instead from relative ampleness of $-E$; curves avoiding the affine chart do not meet E . For large $a > 1$, $aA - E$ is ample. The identity

$$A - E = \frac{a - 1}{a(1 + \sigma_0) - 1} (A - (1 + \sigma_0)E) + \frac{\sigma_0}{a(1 + \sigma_0) - 1} (aA - E)$$

therefore makes $A - E$ ample; see [3, Corollary 1.4.10]. These statements do not require a smooth or normal blow-up.

The large-degree graded-piece and vanishing theorems for the Rees algebra give

$$p_*\mathcal{O}_{X'}(-nE) = I^n, \quad R^q p_*\mathcal{O}_{X'}(-nE) = 0 \quad (q > 0, n \text{ sufficiently large}).$$

These are ordinary powers, not integral closures. The relevant general facts are [10, Tags 0AG6 and 0AG7]. Projection, Leray, and Serre vanishing now give

$$H^1(X, I^n \otimes L^n) = H^1(X', \mathcal{O}_{X'}(n(A - E))) = 0$$

for large n . Hence global sections of L^n realize every local class modulo I^n . Each generator of I has weight R , so

$$I^n \subseteq \{\text{series of weight at least } nR\}.$$

Surjectivity onto the former quotient therefore implies surjectivity onto the desired smaller coefficient quotient. Completion does not change the finite-length local quotients. Finally, for sufficiently large n , global sections of L^n are restrictions of homogeneous polynomials of degree n in the embedding coordinates, by Serre vanishing for the projective ideal sheaf. They have W -degree at most nR on the affine chart. Taking $H = nR$ and undoing the triangular logarithmic substitution proves (3.2). $\square$

4 The algebraic determinant and its arithmetic bound

Fix $\nu > 2\kappa$ and assume (2.1). For now take finite parameters m, K, w_0, v_0, θ satisfying the interpolation hypotheses, and choose good approximants p_i/q_i , $q_i \geq 2$, $p_i \neq 0$, with separated $w_i = \lceil \log q_i \rceil$. Their precise order of choice is given in Section 6. Let $F_0 > 2/\theta$, and set

$$T_i = \left\lceil \frac{F_0 w_i}{v_0} \right\rceil, \quad G_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1}}{k} t^k, \quad L_i = \text{lcm}(1, \dots, T_i). \quad (4.1)$$

Using T_i , rather than $T_i - 1$, in the least common multiple is harmless and agrees with the formal arithmetic bound. The omitted tail has weight at least $v_0 T_i \geq F_0 w_i > w_i/\theta$.

4.1 Matrix, rank, and one minor over the number field

Index columns by $(h, \alpha) \in \mathbb{N} \times \mathbb{N}^m$ satisfying $w_0 h + w \alpha \leq H$, and rows by (j, ℓ, β) satisfying

$$0 \leq j < K, \quad v_0 \ell + w \beta / \theta < H.$$

Define the matrix over F by

$$M_H[(j, \ell, \beta), (h, \alpha)] = [t^\ell u^\beta] (\xi^j (1+t))^h \prod_{i=1}^m (j \gamma p_i / q_i + G_i(t) + u_i)^{\alpha_i}. \quad (4.2)$$

Explicitly, for $\beta \leq \alpha$, its entry is

$$\xi^{jh} \binom{\alpha}{\beta} [t^\ell] (1+t)^h \prod_i (j \gamma p_i / q_i + G_i(t))^{\alpha_i - \beta_i}; \quad (4.3)$$

it is zero otherwise.

Fix one embedding in S . If the powers of ξ are distinct, the centers

$$(\sigma(\xi)^j, (j \sigma(\gamma) p_i / q_i)_{i=1}^m)$$

satisfy case (i) of Theorem 3.1. If $\xi = 1$, each additive coordinate is injective in j because $\gamma p_i/q_i \neq 0$, and case (ii) applies. The truncation in (4.1) preserves the jet quotient. It follows that, for sufficiently large divisible H , the complex image of (4.2) has full row rank. Select a square minor using every row, and denote its determinant by $\Delta_H \in F^\times$ and its size by $M = M(H)$. The chosen columns may depend on H . Every embedding sends this same determinant to the determinant of the corresponding complex minor; none of these images is zero. We never choose different minors at different embeddings.

The fixed-weight simplex counts give

$$M(H) \sim \frac{K\theta^m H^{m+1}}{(m+1)!v_0 \prod_i w_i}. \quad (4.4)$$

Define

$$\bar{b}_H = \frac{1}{MH} \sum_{(j,\ell,\beta) \text{ row}} w\beta, \quad 0 \leq \bar{b}_H \leq \theta. \quad (4.5)$$

The simplex asymptotic follows, for example, by comparing unit cubes with the bounded shifts of the corresponding real simplex. Changing a weak boundary inequality to a strict one does not change its leading volume.

4.2 Clearing the base, slope, and logarithmic denominators

Choose positive integers D_ξ, D_γ such that $D_\xi \xi, D_\gamma \gamma \in \mathcal{O}_F$, and put

$$\mathcal{D}_H = D_\xi^{K\lfloor H/w_0 \rfloor} \prod_{i=1}^m (D_\gamma L_i)^{\lfloor H/w_i \rfloor}. \quad (4.6)$$

Scale each column by $\prod_i q_i^{\alpha_i}$, each row by $\prod_i q_i^{-\beta_i}$, and every entry by $\mathcal{D}_H$.

Lemma 4.1. *Every scaled entry belongs to $\mathcal{O}_F$. Hence the scaled determinant is a nonzero algebraic integer*

$$Z_H = S_H \Delta_H, \quad S_H = \mathcal{D}_H^M \prod_{\text{selected columns } i} \prod_i q_i^{\alpha_i} \prod_{\text{rows } i} q_i^{-\beta_i} > 0.$$

Here $S_H \in \mathbb{Q}$, and it need not be an integer.

Proof. Entries with $\beta \not\leq \alpha$ vanish. For the others, the row and column factors turn (4.3) into

$$\xi^{jh} \binom{\alpha}{\beta} [t^\ell] (1+t)^h \prod_i (j\gamma p_i + q_i G_i(t))^{\alpha_i - \beta_i}.$$

The polynomial $D_\gamma L_i(j\gamma p_i + q_i G_i(t))$ has integral coefficients. Its required power is at most $\alpha_i \leq \lfloor H/w_i \rfloor$. Also $jh \leq K\lfloor H/w_0 \rfloor$, so

$$D_\xi^{K\lfloor H/w_0 \rfloor} \xi^{jh} = (D_\xi \xi)^{jh} D_\xi^{K\lfloor H/w_0 \rfloor - jh}$$

is integral. This proves entrywise integrality. The determinant of an integral matrix is integral, and the scalings are nonzero. $\square$

Let $\Lambda = \log 4 + 4$. We use the elementary bound $\log \text{lcm}(1, \dots, n) \leq \Lambda n$. Indeed, the elementary Chebyshev bound $\vartheta(u) \leq u \log 4$, obtained from binomial coefficients, and the prime-power identity for ψ give, for $n \geq 1$,

$$\log \text{lcm}(1, \dots, n) = \psi(n) \leq n \log 4 + 2\sqrt{n} \log n \leq (\log 4 + 4)n.$$

The last step uses $\log n \leq 2\sqrt{n}$; $n = 0$ is immediate. This deliberately loose constant is exactly `OAI.PiExponent.Arithmetic.lcmConstant` in the pinned source. The sharper constant in the informal argument of [8, Section 3.1] is not needed. Since $T_i \leq F_0 w_i / v_0 + 1$,

$$\frac{\log \mathcal{D}_H}{H} \leq \frac{\Lambda F_0 m}{v_0} + (\Lambda + \log D_\gamma) \sum_i \frac{1}{w_i} + \frac{K \log D_\xi}{w_0}. \quad (4.7)$$

Write $w_* = \min_i w_i$. For every selected column, $\sum_i \alpha_i \log q_i \leq H$. For a row, $\log q_i \geq w_i - 1$ and $|\beta| \leq \theta H / w_*$, so

$$\sum_i \beta_i \log q_i \geq w\beta - \theta H / w_*.$$

Consequently

$$\frac{\log S_H}{MH} \leq 1 - \bar{b}_H + E_{\text{ar}}^0, \quad E_{\text{ar}}^0 = \frac{\Lambda F_0 m}{v_0} + (\Lambda + \log D_\gamma) \sum_i \frac{1}{w_i} + \frac{K \log D_\xi}{w_0} + \frac{\theta}{w_*}. \quad (4.8)$$

Proposition 4.2 (Norm lower bound). *Set*

$$\mathcal{N}_H = \frac{1}{sMH} \sum_{\sigma: F \hookrightarrow \mathbb{C}} \log |\sigma(\Delta_H)|, \quad E_{\text{ar}} = \kappa E_{\text{ar}}^0.$$

Then

$$\mathcal{N}_H \geq -\kappa(1 - \bar{b}_H) - E_{\text{ar}}. \quad (4.9)$$

Proof. The norm of the nonzero algebraic integer Z_H is a nonzero rational integer. Therefore

$$1 \leq |N_{F/\mathbb{Q}}(Z_H)| = S_H^d \prod_{\sigma: F \hookrightarrow \mathbb{C}} |\sigma(\Delta_H)|.$$

Take logarithms, divide by sMH , and apply (4.8). $\square$

For $x = \log 3$, take $F = \mathbb{Q}$, $\gamma = 1$, $\xi = 3$, and $D_\xi = D_\gamma = 1$: the factor 3^{jh} is integral and costs no denominator. For a positive rational $r = a/b$, one can take $D_\xi = b$; the extra cost is at most $K \log b / w_0$, which the parameter choice will make small.

5 Analytic bounds at all embeddings

Define fixed size constants

$$\Gamma = \max \left(1, \max_\sigma |\sigma(\gamma)| \right), \quad \Xi = \max \left(1, \max_\sigma |\sigma(\xi)| \right), \quad B_0 = \Gamma(|x| + 1), \quad R_0 = 100(1 + \Gamma|x|).$$

The maxima range over all embeddings, not just S . Thus $B_0 \geq 1$, $R_0 \geq 100$. Put $R = R_0 K$.

5.1 An exact row translation

For a polynomial column P and $a \in \mathbb{N}^m$, consider the entire function

$$f_{a,P}(z) = [u^a] P(e^z, z + u_1, \dots, z + u_m). \quad (5.1)$$

For $P = Y^h X^\alpha$, this is $\binom{\alpha}{a} e^{hz} z^{|\alpha| - |a|}$ when $a \leq \alpha$, and zero otherwise. In particular these functions do not depend on the center index j .

Fix $\sigma \in S$ and put

$$\lambda_\sigma = \sigma(\gamma)x, \quad z = j\lambda_\sigma + \log(1+t), \quad \epsilon_i = j\sigma(\gamma)(p_i/q_i - x), \quad \tau_i(t) = G_i(t) - \log(1+t).$$

The two identities

$$e^z = \sigma(\xi)^j(1+t), \quad j\sigma(\gamma)p_i/q_i + G_i(t) + u_i = z + u_i + \epsilon_i + \tau_i(t)$$

give the exact row-vector expansion

$$\text{row}_{j,\ell,\beta} = \sum_{\substack{a \geq \beta \\ wa \leq H}} \binom{a}{\beta} \sum_{0 \leq e \leq a-\beta} \binom{a-\beta}{e} \epsilon^{a-\beta-e} \sum_{k=0}^{\ell} \left([t^k] \prod_i \tau_i(t)^{e_i} \right) \left([t^{\ell-k}] f_{a,P}(j\lambda_\sigma + \log(1+t)) \right)_P. \quad (5.2)$$

The scalar multiplying a translated row is independent of the column P . This is what permits determinant multilinearity. The identity is an equality of finite coefficient calculations: the column space bounds a , and only Taylor coefficients through ℓ occur.

Lemma 5.1 (Scalar cost). *Every scalar in (5.2) has absolute value at most*

$$\exp\{-\nu w(a-\beta) + HE_{\text{tr}}\}, \quad E_{\text{tr}} = \frac{\nu}{F_0} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2K\Gamma) + \nu}{w_*}. \quad (5.3)$$

Each row has at most

$$Q_H = (\lfloor H \rfloor + 1)^{2m} (\lfloor H/v_0 \rfloor + 1)$$

choices in this expansion.

Proof. The approximation and ceiling inequalities give

$$|\epsilon_i| \leq K\Gamma q_i^{-\nu} \leq K\Gamma e^\nu e^{-\nu w_i}.$$

Each τ_i starts at degree T_i . A nonzero coefficient in its product requires $\sum_i e_i T_i \leq k \leq \ell < H/v_0$, whence $we < H/F_0$. On $|t| = 1/2$, the logarithmic tails have absolute value at most $\sum_{n \geq 1} 2^{-n}/n = \log 2 < 1$. Cauchy's estimate bounds the coefficient of their product by 2^k . The binomial coefficients contribute at most $4^{|a|}$, and $|a| \leq H/w_*$. In the approximation factor, replacing $w(a-\beta-e)$ by $w(a-\beta)$ costs at most $\nu H/F_0$. These inequalities imply (5.3), with the harmless extra factor 2 inside $\log(2K\Gamma)$. Finally $w_i \geq 1$, so both a and e have coordinates at most $\lfloor H \rfloor$; $k \leq \lfloor H/v_0 \rfloor$. $\square$

5.2 The repeated-coefficient saving

Select one translated row from each original row and omit its scalar. Call the resulting square matrix T . Let n_a count the chosen rows with transverse index a , so $\sum_a n_a = M$. Put

$$c = \frac{\log 2}{4}, \quad E_{\text{hol}} = \frac{R_0 K}{w_0} + \frac{\log 2}{v_0} + \frac{\log(2R_0 K)}{w_*}.$$

Lemma 5.2 (Collision estimate). *Uniformly over selected columns, translated-row choices, and $\sigma \in S$,*

$$|\det T| \leq \exp \left\{ -c \sum_a n_a^2 + MH(E_{\text{hol}} + \rho_H) \right\}, \quad \rho_H \rightarrow 0. \quad (5.4)$$

One may take

$$\rho_H = \frac{\log M + c - \log(1 - 2^{-1/2})}{H}.$$

Proof. On $|z| \leq R$, the column degree bound gives

$$|f_{a,P}(z)| \leq \exp \left\{ H \left(\frac{R}{w_0} + \frac{\log(2R)}{w_*} \right) \right\} =: \mathcal{B}.$$

On $|t| = 1/2$,

$$|j\lambda_\sigma + \log(1+t)| \leq (K-1)\Gamma|x| + \log 2 < R/2.$$

Expand $f_{a,P}(z) = \sum_{d \geq 0} b_{a,P,d}(z/R)^d$. Cauchy's estimate gives $|b_{a,P,d}| \leq \mathcal{B}$. The Taylor test of order r applied to $(z/R)^d$ has absolute value at most $2^r 2^{-d}$.

If two rows use the same pair (a, d) , their coefficient vectors $(b_{a,P,d})_P$ agree, and that determinant term is zero. For a nonzero term the degrees in each a -group are distinct; therefore

$$\sum_{\text{rows}} d \geq \sum_a \binom{n_a}{2} = \frac{\sum_a n_a^2 - M}{2}.$$

Use half of the factor $2^{-\sum d}$ for this lower bound and sum its remaining half over all nonnegative degrees independently. The latter sum is at most $(1 - 2^{-1/2})^{-M}$. The determinant of each coefficient matrix is at most $M! \mathcal{B}^M$, and the sum of the Taylor-test orders is at most MH/v_0 . Hence

$$|\det T| \leq M! \mathcal{B}^M 2^{MH/v_0} \exp\{-c(\sum_a n_a^2 - M)\}(1 - 2^{-1/2})^{-M}.$$

Use $M! \leq M^M$ to obtain the asserted expression. The series manipulations are justified by the same convergent geometric majorant. Equation (4.4) shows $\log M/H \rightarrow 0$. This proves the stated uniformity as well. $\square$

The argument is the coefficient-collision mechanism of [8, Lemma 3.3]. The change to real logarithmic centers requires only the new radius bound; it does not remove the repeated coefficient rows responsible for the saving.

5.3 Two alternatives for the full determinant

Fix $0 < A < 1$ and $0 < \eta < 1$, and let

$$D_A(H) = \#\{a \in \mathbb{N}^m : wa \leq AH\}, \quad L_H = \frac{\eta^2 M}{H D_A(H)}.$$

The simplex counts imply

$$D_A(H) \sim \frac{A^m H^m}{m! \prod_i w_i}, \quad L_H \longrightarrow L = \frac{\eta^2 K \theta^m}{(m+1)v_0 A^m}. \quad (5.5)$$

If at least ηM chosen rows have $wa \leq AH$, Cauchy–Schwarz gives $\sum_a n_a^2 \geq \eta^2 M^2 / D_A(H) = MHL_H$. For a nonzero scalar, $a \geq \beta$; thus the approximation exponent in (5.3) is nonpositive and may be discarded. If fewer than ηM rows have $wa \leq AH$, then

$$\sum_{\text{rows}} w(a - \beta) > MH(A(1 - \eta) - \bar{b}_H).$$

In this alternative discard the nonpositive collision exponent. The determinant expansion has at most Q_H^M terms. Taking logarithms of the triangle inequality yields, at every compatible embedding,

$$\frac{\log |\sigma(\Delta_H)|}{MH} \leq E_{\text{tr}} + E_{\text{hol}} + \rho_H + \frac{\log Q_H}{H} + \max\{-cL_H, -\nu(A(1 - \eta) - \bar{b}_H)\}. \quad (5.6)$$

This upper bound applies to every selected nonzero full-row minor. It is uniform in its selected columns; no analytic choice of a special minor is made after the arithmetic choice.

5.4 The other embeddings

For an arbitrary embedding σ , use (4.3) directly. On $|t| = 1/2$,

$$|j\sigma(\gamma)p_i/q_i| \leq KB_0, \quad |G_i(t)| < 1.$$

The binomial coefficient costs at most $2^{|\alpha|}$; $|\sigma(\xi)^{jh}| \leq \Xi^{Kh}$; $|1 + t|^h \leq (3/2)^h$; and coefficient extraction costs 2^ℓ . It follows that every entry is at most $\exp(HU)$, where the convenient common bound used by the formal proof is

$$U = \frac{(\log \Xi + \log(3/2))K}{w_0} + \frac{\log(2(KB_0 + 2))}{w_*} + \frac{\log 2}{v_0} \geq 0. \quad (5.7)$$

Thus

$$\frac{\log |\sigma(\Delta_H)|}{MH} \leq U + \frac{\log M}{H}. \quad (5.8)$$

We claim no collision saving at incompatible embeddings.

Proposition 5.3 (All-embedding upper bound). *Let*

$$E_{\text{an}} = E_{\text{tr}} + E_{\text{hol}} + \kappa U.$$

For the same minor used in Proposition 4.2,

$$\mathcal{N}_H \leq E_{\text{an}} + o(1) + \max\{-cL_H, -\nu(A(1 - \eta) - \bar{b}_H)\}. \quad (5.9)$$

The remainder tends to zero for the fixed data and is uniform over the selected columns.

Proof. Sum (5.6) over S , and (5.8) over its complement, then divide by s . Since $U + \log M/H \geq 0$, replacing the complement size by d only enlarges the upper bound. One resulting remainder is

$$\rho_H + \frac{\log Q_H}{H} + \kappa \frac{\log M}{H},$$

which tends to zero by the polynomial growth of M and Q_H . $\square$

6 Parameter choices and the final contradiction

Lemma 6.1 (The comparison gap). *Let $\kappa \geq 1$ and $\nu > 2\kappa$. There are positive rational numbers θ, A, B, C and a real $\eta \in (0, 1)$ such that*

$$0 < \theta < A < B < 1, \quad C > 1, \quad C\theta < 1, \quad CB < 1, \quad C\theta/B > 1, \quad B/A > 1, \quad (6.1)$$

$$g := \nu(A(1 - \eta) - \theta) - \kappa(1 - \theta) > 0. \quad (6.2)$$

Proof. Choose a rational b_0 with

$$\frac{1}{2} < b_0 < 1 - \frac{\kappa}{\nu}.$$

For a sufficiently small positive rational δ , put $\theta = 1 - \delta$, $A = 1 - b_0\delta$. Then $0 < \theta < A < 1$,

$$\nu(A - \theta) - \kappa(1 - \theta) = (\nu(1 - b_0) - \kappa)\delta > 0,$$

and $\theta - A^2 = (2b_0 - 1)\delta - b_0^2\delta^2 > 0$. Choose rationals

$$A/\theta < C < 1/A, \quad A < B < \min(1/C, C\theta).$$

The intervals are nonempty because $A^2 < \theta$. Also $C\theta < \theta/A < 1$, giving all the ratio inequalities. Finally choose $\eta > 0$ small enough to preserve the positive gap. $\square$

The constants in the preceding estimates can all be made small in the required order. To see this without suppressing any number-field cost, put $E = E_{\text{ar}} + E_{\text{an}}$ and define

$$\begin{aligned}\mathcal{C} &= R_0 + \kappa \log D_\xi + \kappa(\log \Xi + \log(3/2)), \\ \mathcal{W} &= \kappa\theta + \log 4 + \log(2KT) + \nu + \log(2R_0K) + \kappa \log(2(KB_0 + 2)).\end{aligned}$$

All these quantities are nonnegative. The exact error identity is

$$\begin{aligned}E &= \frac{\nu}{F_0} + \frac{\kappa\Lambda F_0 m + (\kappa + 2) \log 2}{v_0} + \mathcal{C} \frac{K}{w_0} \\ &\quad + \kappa(\Lambda + \log D_\gamma) \sum_{i=1}^m \frac{1}{w_i} + \frac{\mathcal{W}}{w_*}.\end{aligned}\tag{6.3}$$

In particular, algebraic denominators and unconstrained conjugates appear explicitly rather than being hidden in the $o(1)$ in H .

Proposition 6.2 (Admissible finite choices). *Assuming (2.1), all fixed data can be selected so that the interpolation theorem applies and*

$$E < g, \quad cL > \kappa + E.\tag{6.4}$$

Proof. Choose θ, A, B, C, η by Lemma 6.1, and set $\epsilon_0 = \min(g/2, 1/2) > 0$. First choose $F_0 > 2/\theta$ such that $\nu/F_0 < \epsilon_0/3$. For an integer $m \geq 1$, define

$$K = \lfloor C^m \rfloor, \quad w_0 = B^{-m}, \quad v_0 = 2K\theta^m w_0.\tag{6.5}$$

They satisfy

$$K(w_0/v_0)\theta^m = \frac{1}{2}, \quad K\theta^m \leq (C\theta)^m < 1.$$

The second inequality allows the common-fiber branch as well as the distinct-center branch. Moreover

$$\frac{K}{w_0} \leq (CB)^m \longrightarrow 0, \quad v_0 \sim 2(C\theta/B)^m \longrightarrow \infty, \quad \frac{m}{v_0} \longrightarrow 0.$$

Substitution in (5.5) gives the exact expression

$$L = \frac{\eta^2}{2(m+1)} (B/A)^m \longrightarrow \infty.\tag{6.6}$$

Thus one can fix a finite m large enough that

$$\frac{\kappa\Lambda F_0 m + (\kappa + 2) \log 2}{v_0} + \mathcal{C} \frac{K}{w_0} < \epsilon_0/3, \quad cL > \kappa + 1.$$

This fixes m, K, w_0, v_0 , and hence $\mathcal{W}$.

Now choose σ_0 and the weight thresholds for Theorem 3.1, imposing also its common-fiber volume inequality. Choose the finitely many approximants p_i/q_i successively from (2.1). Their weights can satisfy every separation threshold and can all exceed a common bound large enough that

$$\kappa(\Lambda + \log D_\gamma) \sum_i \frac{1}{w_i} + \frac{\mathcal{W}}{w_*} < \epsilon_0/3.$$

Increase the denominator thresholds as needed to ensure $q_i \geq 2$ and $p_i \neq 0$. This step is valid because the interpolation weight thresholds are independent of the centers, which themselves involve p_i/q_i .

Equation (6.3) now gives $E < \epsilon_0 < g$; also $cL > \kappa + 1 > \kappa + E$. Every choice is finite. Only after fixing this entire list do we let the sufficiently divisible parameter H increase. $\square$

Proof of Theorem 1.2. Normalize (γ, ξ) by Lemma 2.3. Fix $\nu > 2\kappa$ and suppose (2.1). Make the choices of Proposition 6.2. For arbitrarily large divisible H , interpolation supplies a nonzero selected determinant. Propositions 4.2 and 5.3 then give

$$-\kappa(1 - \bar{b}_H) - E_{\text{ar}} \leq E_{\text{an}} + o(1) + \max\{-cL_H, -\nu(A(1 - \eta) - \bar{b}_H)\}.$$

Both entries of this maximum are incompatible with the lower bound for sufficiently large H . For the collision entry, $0 \leq \bar{b}_H$ implies $\kappa(1 - \bar{b}_H) \leq \kappa$, whereas $cL_H \rightarrow cL > \kappa + E$. For the other entry, since $\nu > \kappa$ and $\bar{b}_H \leq \theta$,

$$\nu(A(1 - \eta) - \bar{b}_H) - \kappa(1 - \bar{b}_H) \geq \nu(A(1 - \eta) - \theta) - \kappa(1 - \theta) = g > E.$$

The strict margins absorb the common remainder $o(1)$. This contradiction excludes (2.1). It proves $\text{ELB}(x, 2\kappa)$, including the bound for unreduced rational representations. Lemma 2.2 proves irrationality and the exponent bounds. The rational-logarithm theorem and algebraic-base corollary follow from the specializations already given. $\square$

Remark 6.3 (What the exponent threshold means). The condition $\nu > 2\kappa$ is used to choose $1/2 < b_0 < 1 - \kappa/\nu$. It supplies the two simultaneous requirements $A^2 < \theta$ and a positive determinant gap. This is the precise source of $2d/s$. At the boundary $\nu = 2\kappa$, this argument gives neither that interval nor a strict gap. For rational logarithms, exponent two means the eventual inequalities for every $2 + \varepsilon$; it does not assert an eventual lower bound with the exact exponent 2.

7 Special cases, comparison with the π proof, and limitations

7.1 Rational powers and rational affine changes

Lemma 7.1. *If $b \geq 2$, $\text{ELB}(x, b)$, and $a, c \in \mathbb{Q}$ with $a \neq 0$, then $\text{ELB}(ax + c, b)$.*

Proof. For a fixed $\nu > b$, choose $b < \nu' < \nu$. Write $a = A/B$, $c = C/D$, with $B, D > 0$. For any p/q , the rational number $(p/q - c)/a$ has an integer representation with denominator $|A|Dq$. Consequently, for sufficiently large q ,

$$|ax + c - p/q| \geq |a|(|A|Dq)^{-\nu'}.$$

The fixed positive factor $|a|(|A|D)^{-\nu'}$ is eventually at least $q^{-(\nu-\nu')}$, proving the assertion. $\square$

Corollary 7.2. *Suppose $\alpha > 0$, $n \in \mathbb{N}_{>0}$, and $\alpha^n = r \in \mathbb{Q}_{>0} \setminus \{1\}$. Then*

$$\mu(\log \alpha) = 2.$$

In particular, $\mu(\log \sqrt{2}) = 2$.

Proof. The identity $\log \alpha = (1/n) \log r$, together with Theorem 1.1, Lemma 7.1, and Lemma 2.2, proves the result. $\square$

This sharp conclusion is stronger than the degree bound in this particular family. The proof uses rational scaling, not invariance under arbitrary algebraic scaling.

7.2 Arctangents of rational numbers

Corollary 7.3. *For every $r \in \mathbb{Q} \setminus \{0\}$,*

$$\mu(\arctan r) = 2.$$

Proof. Use $F = \mathbb{Q}(i)$, $\gamma = 2i$, $\xi = (1 + ir)/(1 - ir)$, and $x = \arctan r \neq 0$. The identity

$$e^{2i \arctan r} = \frac{1 + ir}{1 - ir}$$

establishes compatibility for the given embedding. Its complex conjugate establishes compatibility for the other embedding. Thus $s = d = 2$. For $r = 1$, $\xi = i$ is a root of unity; the root-of-unity branch of the theorem is genuinely needed here. $\square$

7.3 Quadratic-normalized logarithms

Corollary 7.4. *Let $a, b, D \in \mathbb{Q}$, $D > 0$, and suppose*

$$a^2 - b^2 D = 1, \quad \alpha = a + b\sqrt{D} > 1.$$

Then

$$\mu\left(\frac{\log(a + b\sqrt{D})}{\sqrt{D}}\right) = 2.$$

In particular,

$$\mu\left(\frac{\log(3 + 2\sqrt{2})}{\sqrt{2}}\right) = 2. \tag{7.1}$$

Proof. Take $F = \mathbb{Q}(\sqrt{D})$, $\gamma = \sqrt{D}$, $\xi = a + b\sqrt{D}$, and $x = \log \alpha / \sqrt{D} \neq 0$. Every embedding sends $\sqrt{D}$ to either $\sqrt{D}$ or $-\sqrt{D}$. The first choice gives $e^{\sqrt{D}x} = \alpha$. The norm identity gives $a - b\sqrt{D} = \alpha^{-1}$, so the second choice gives

$$e^{-\sqrt{D}x} = \alpha^{-1} = a - b\sqrt{D}.$$

Every embedding is compatible, whether the field has degree one or two. $\square$

The normalization in this corollary is essential to the asserted specialization. It does not imply $\mu(\log(3 + 2\sqrt{2})) = 2$ by removing the factor $\sqrt{2}$. The general algebraic-base result gives the separate bound $2 \leq \mu(\log(3 + 2\sqrt{2})) \leq 4$.

7.4 How the argument differs from the original π proof

The relation to [8] is at the level of proved machinery, not a substitution into its final theorem. The following comparison fixes the main changes.

Feature	Original π construction	Logarithmic extension
Exponential identity	$e^{2i\pi} = 1$	$e^{\log r} = r$, or $e^{\sigma(\gamma)x} = \sigma(\xi)$
Multiplicative centers	Common fiber $Y = 1$	Distinct powers r^j or ξ^j ; common fiber after root-of-unity reduction

Feature	Original π construction	Logarithmic extension
Local coordinates	$t = Y - 1$	$t = Y/y_j - 1$; the same logarithmic frame
Arithmetic	Gaussian-integer norm after denominator clearing	Rational integers for rational logarithms; an algebraic integer and its full field norm in general
Analytic saving	Repeated (a, d) coefficient rows	The same repetition argument at every compatible embedding; direct growth bounds at the others
Final coefficient	1 in the normalized arithmetic comparison	d/s , giving threshold $2d/s$

For π , the compatible data may be viewed as $F = \mathbb{Q}(i)$, $\gamma = 2i$, $\xi = 1$, $x = \pi$: both embeddings are compatible. For $\log r$, the field is $\mathbb{Q}$, the slope is 1, and the real centers require the distinct-fiber argument. The analytic translation is preserved exactly, while the radius constants change. The proof of the new interpolation theorem, the identification with (4.2), the field-denominator estimates, and the estimates at all embeddings are indispensable additions.

7.5 Why no theorem for all positive real bases is possible

Proposition 7.5. *The map $\alpha \mapsto \log \alpha$ on positive real numbers is surjective onto $\mathbb{R}$. There is a positive real $\alpha \neq 1$ with rational logarithm, and there is such an α whose irrational logarithm has no finite irrationality exponent.*

Proof. For every $y \in \mathbb{R}$, $\alpha = e^y > 0$ satisfies $\log \alpha = y$. Taking $y = 1$ gives the rational example. For the second, let

$$L = \sum_{n=0}^{\infty} 10^{-n!}, \quad \alpha = e^L.$$

The standard Liouville argument applies: truncation through N has denominator $q_N = 10^{N!}$, while its positive tail is at most $2q_N^{-(N+1)}$. These approximations prove that L is an irrational Liouville number and admits no eventual lower bound of any finite exponent. Thus $\log \alpha = L$ has the claimed property. $\square$

These examples concern arbitrary real inputs. They are not counterexamples to the rational- or algebraic-base theorems. The formal statements express the second conclusion as failure of $\text{ELB}(\log \alpha, b)$ for every finite real b , avoiding any ambiguous convention for a real-valued supremum of an unbounded set.

8 Formal verification and reproducibility

8.1 What the formal endpoints assert

The Lean definition `LogarithmExtension.EventualLowerBound` is exactly the predicate ELB in Section 2; it quantifies over every integer numerator and every sufficiently large natural denominator. The main declarations include

```
logThree_eventual_lower_bound
logThree_irrationalityExponent_eq_two
rationalLog_eventual_lower_bound
```

```

rationalLog_irrationalityExponent_eq_two
algebraicLog_irrationalityExponent_bounds
CompatibleLog.compatible_embedding_eventual_lower_bound
CompatibleLog.compatible_embedding_irrationalityExponent_bounds
CompatibleLog.all_embeddings_irrationalityExponent_eq_two

```

All are in the namespace `LogarithmExtension`. The unbundled compatible-embedding theorem takes only the number field, γ, ξ, x , their stated nonzero conditions, a nonempty finite set of embeddings, and the exponential identities (1.2). It has no interpolation, nonvanishing, analytic-bound, or irrationality premise. The bundled data structure `CompatibleLog.Base` contains these same mathematical data, not a hidden proof of the conclusion.

The names `LogThreeTarget`, `RationalLogarithmTarget`, and `AlgebraicLogarithmTarget` are proposition definitions. They are not axioms: the corresponding `target` declarations prove them.

The exponent definition reused from `OAI.PiExponent` is a real supremum of the positive exponents admitting infinitely many good reduced rational approximants. For every finite-bound endpoint, the code first proves that this set is bounded above, and that it contains 2. The resulting supremum therefore agrees with the usual extended-real definition in this paper. The Liouville counterexample is expressed as absence of any finite eventual bound, rather than as a possibly misleading real supremum.

8.2 Correspondence of arguments and source files

All extension paths below are relative to `LogarithmExtension/`. The earlier dedicated `LogThree`, `RationalLog`, and `AlgebraicLog` pipelines remain checked; the `CompatibleLog` pipeline supplies the unified theorem proved here.

Mathematical component	Principal extension modules
Actual target and finite exponent semantics	<code>Target.lean</code> , <code>Endpoint.lean</code> , <code>GeneralExponent.lean</code>
Normalized logarithmic coordinates and rigidity	<code>NormalizedLogJet.lean</code> , <code>CoordinateScaling.lean</code> , <code>CurveRigidity.lean</code>
Proposition 3.3, distinct-center case	<code>CurveContactFamily.lean</code> , <code>CurveDerivativeVanishing.lean</code> , <code>CurveInequality.lean</code> , <code>CurveGeometry.lean</code>
Common-fiber branch	<code>CommonFiberCurve.lean</code> , <code>GeometryData.lean</code>
Projective degree, local ideals, and exceptional degree	<code>CurveModelDegrees.lean</code> , <code>NormalizedJetIdeal.lean</code> , <code>CompactNormalizedJetIdeal.lean</code> , <code>LocalContactLength.lean</code> , <code>ExceptionalDegree.lean</code>
Ampleness and global jet generation	<code>BlowupGeometry.lean</code> , <code>BlowupMargin.lean</code> , <code>BlowupAmple.lean</code> , <code>JetAmpleness.lean</code> , <code>GlobalJetPackets.lean</code> , <code>JetSurjectivity.lean</code>
Identification with the actual matrix	<code>BoundedJetMatrix.lean</code> , <code>MatrixIdentification.lean</code> , <code>CompatibleLogGeometry.lean</code>
Equation (5.2) and analytic centers	<code>AnalyticTranslation.lean</code> , <code>CompatibleLogAnalyticCenters.lean</code>
Arithmetic over F , integral norm, and averaging	<code>NumberFieldMatrix.lean</code> , <code>NumberFieldNorm.lean</code> , <code>NumberFieldDeterminant.lean</code> , <code>NumberFieldNormAveraging.lean</code> , <code>CompatibleLogMatrixArithmetic.lean</code>

Mathematical component	Principal extension modules
Actual summands and their finite sum	<code>CompatibleLogLiteralAnalyticSummand.lean</code> , <code>CompatibleLogAnalyticAggregate.lean</code>
Incompatible embeddings and full norm bound	<code>ConjugateAnalyticBound.lean</code> , <code>CompatibleLogAllEmbeddings.lean</code>
Choices and determinant contradiction	<code>CompatibleLogSuccessiveApproximations.lean</code> , <code>CompatibleLogAdmissibleParameters.lean</code> , <code>CompatibleLogDeterminantContradiction.lean</code>
Root-of-unity reduction and final theorem	<code>CompatibleLogBase.lean</code> , <code>CompatibleLogNonTorsion.lean</code> , <code>CompatibleLogMain.lean</code>
Theorem 1.1, including the literal $\log 3$	<code>LogThreeMain.lean</code> , <code>RationalLogMain.lean</code>
Corollaries 1.3–7.4	<code>AlgebraicLogMain.lean</code> , <code>RationalAffine.lean</code> , <code>RationalPowerLog.lean</code> , <code>RationalArctangent.lean</code> , <code>QuadraticNormalizedLog.lean</code>
Proposition 7.5	<code>LogarithmCounterexamples.lean</code>

General results reused from the pinned upstream development include the persistent-component and weighted multiplicity machinery, projective and local algebra, numerical ampleness and vanishing, simplex counts, logarithmic row algebra, collision estimates, and the approximation endpoint lemmas. Theorem $\mu(\pi) = 2$ itself is not the hypothesis from which the logarithm result is deduced. The new target-independent geometry data remove the π -specific approximation hypotheses from the geometric interface.

8.3 Axiom audit and kernel replay

The build and transitive axiom audit use the following commands:

```
lake build -q LogarithmExtension.Verification
lake env lean --trust=0 -Ddebug.skipKernelTC=false \
  LogarithmExtension/Verification.lean
```

The audit checked 1326 extension theorem declarations. Every transitive axiom was one of

`propext`, `Classical.choice`, `Quot.sound`.

It also audited the reference π endpoint separately. A source scan found no proof admissions, new axiom declarations, unsafe or partial declarations, or `native_decide` in the extension. A defined target proposition is not counted as a proof.

The full combined replay was also rerun successfully:

```
lake env lean --run VerifyGeneralLog.lean
```

Its observed results were

```
Replaying 118179 endpoint dependencies in an empty kernel environment.
PASS: all 33 general-logarithm and counterexample endpoints.
PASS: their complete dependency closure rechecked
      in an empty kernel environment.
PASS: only propext, Classical.choice, and Quot.sound allowed as axioms.
```

The selected roots cover $\log 3$, rational and rational-power logarithms, algebraic logarithms, both branches of the compatible theorem, the arctangent and quadratic examples, and the real-input

counterexamples. The checker collects dependencies of both types and values, follows mutual declaration blocks, rejects unsafe or partial declarations and nonstandard axioms, calls `Lean.Replay` in an environment created by `mkEmptyEnvironment`, and checks that the selected roots occur in the replayed environment. This replay uses Lean’s kernel in a fresh environment, not a second, independently implemented proof checker.

8.4 Pinned sources and reproduction

The source package fixes the following revisions:

Component	Version or commit
Lean	4.34.1
Lean commit	5045d0056413266e57c625dcd7c365b10e377c52
Mathlib	d13f23b723b8a846827a245b89c10fc7d3f11612
OpenAI mathematics	adc7f1241b42e322a6451854ab7e4b4c146bf78a

The upstream tracked source was not modified. The ancillary package contains the extension, selected upstream source trees, `lean-toolchain`, `lakefile.toml`, the pinned `lake-manifest.json`, and the replay programs. A SHA-256 inventory identifies the packaged files. Only local source modules required by the logarithm formalization and its verification targets are included.

After installing the specified Lean toolchain, enter the ancillary Lean project directory and run

```
lake exe cache get
lake build LogarithmExtension.Verification
lake env lean --run VerifyGeneralLog.lean
```

Finish the build before starting the replay, and do not rebuild imported modules while it is running. The cache command obtains the Mathlib build cache; compiling its sources is an alternative. External toolchain and package downloads, adequate memory, and disk space are required. The distributed archive contains sources rather than a preinstalled Lean environment.

Acknowledgments

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